SIMBLOCKS // RESULT

2026-10-16 08:03Z

BATTERY FOR A 180 S FIGHT

What is the smallest 6S pack that finishes a full fight?

● PASS VERIFIED

MINIMUM 6S PACK (95% OF FIGHTS) PASS

1500 mAh

at 3.3 V/cell; median fight 1196 mAh

CHARGE USED PASS

SF 1.87

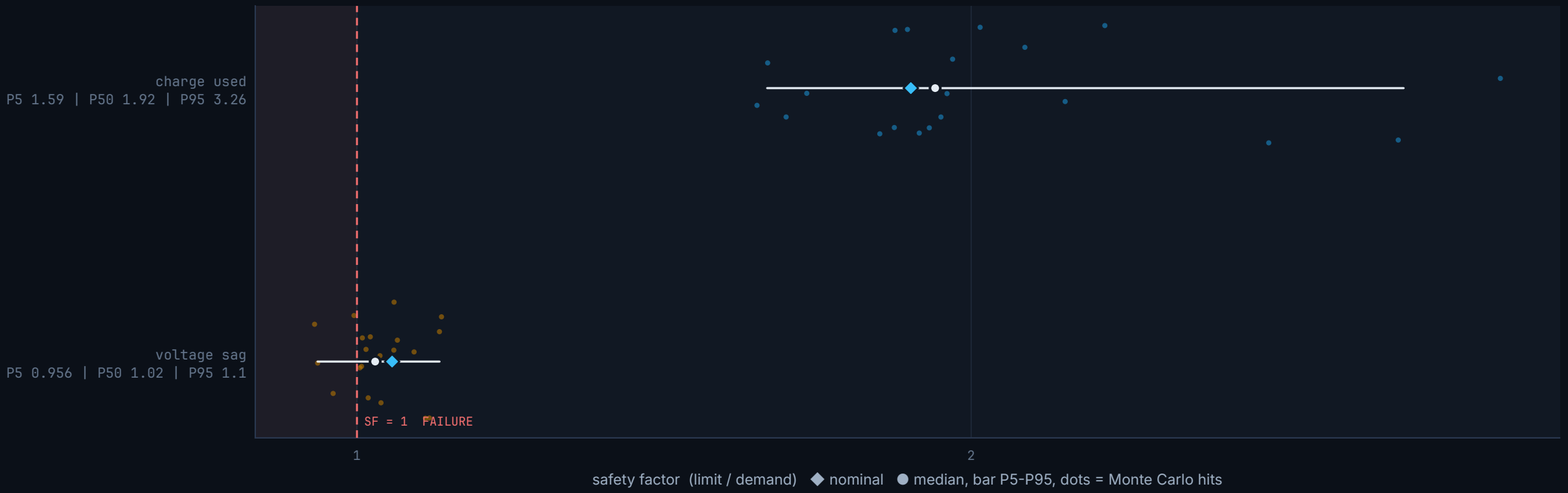
P(fail) 0% | SF P5 1.59

VOLTAGE SAG [GOVERNING] WARN

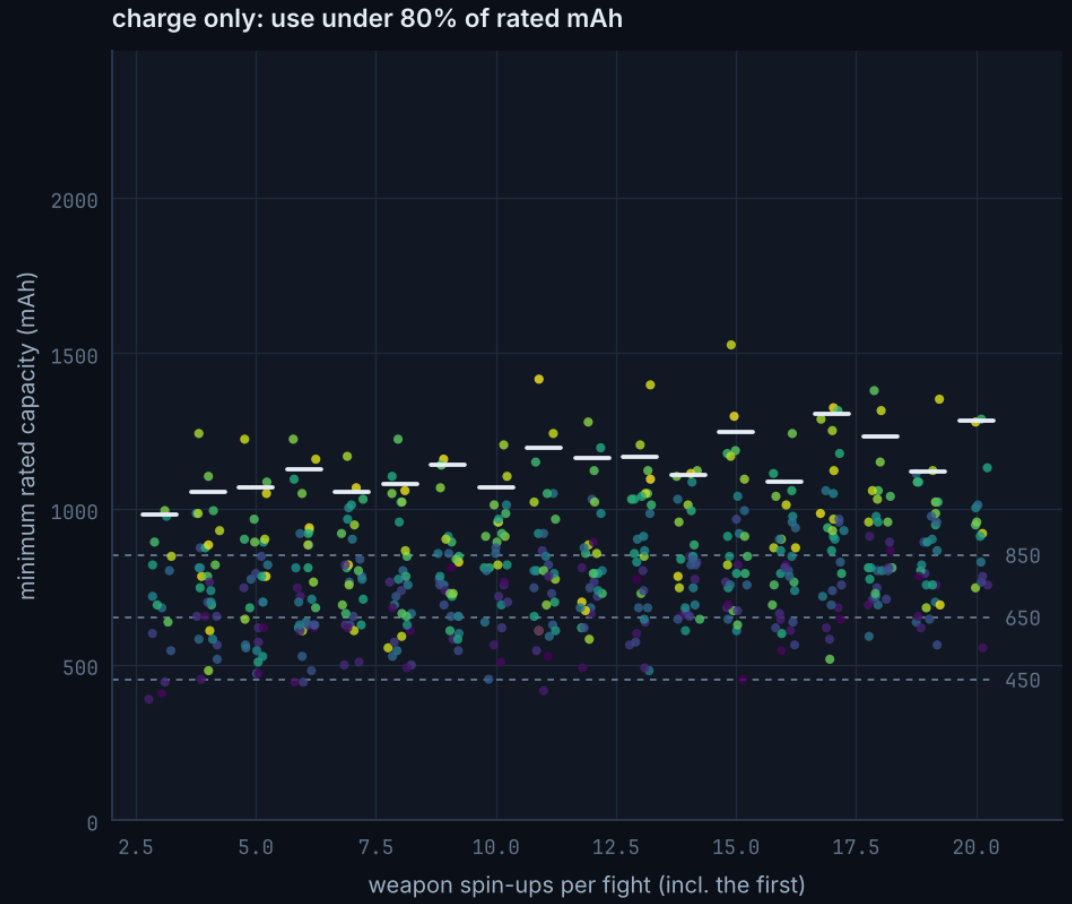
SF 1.04

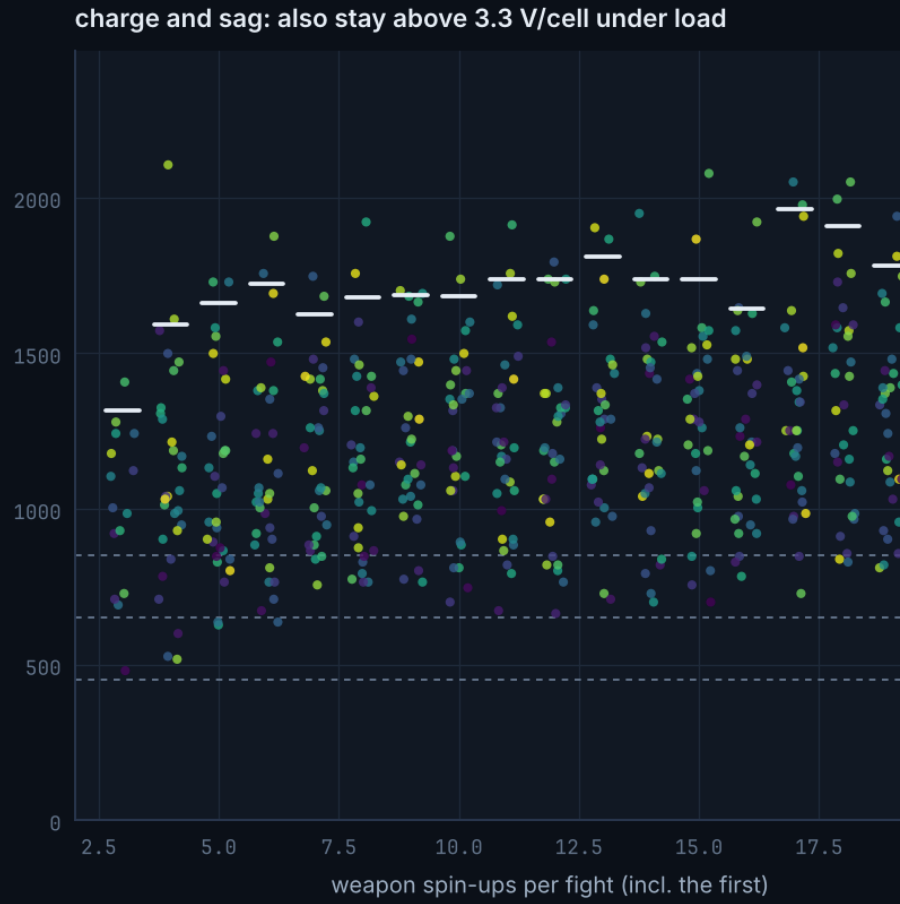
P(fail) 20% | SF P5 0.956

Margin to each user-defined failure mode

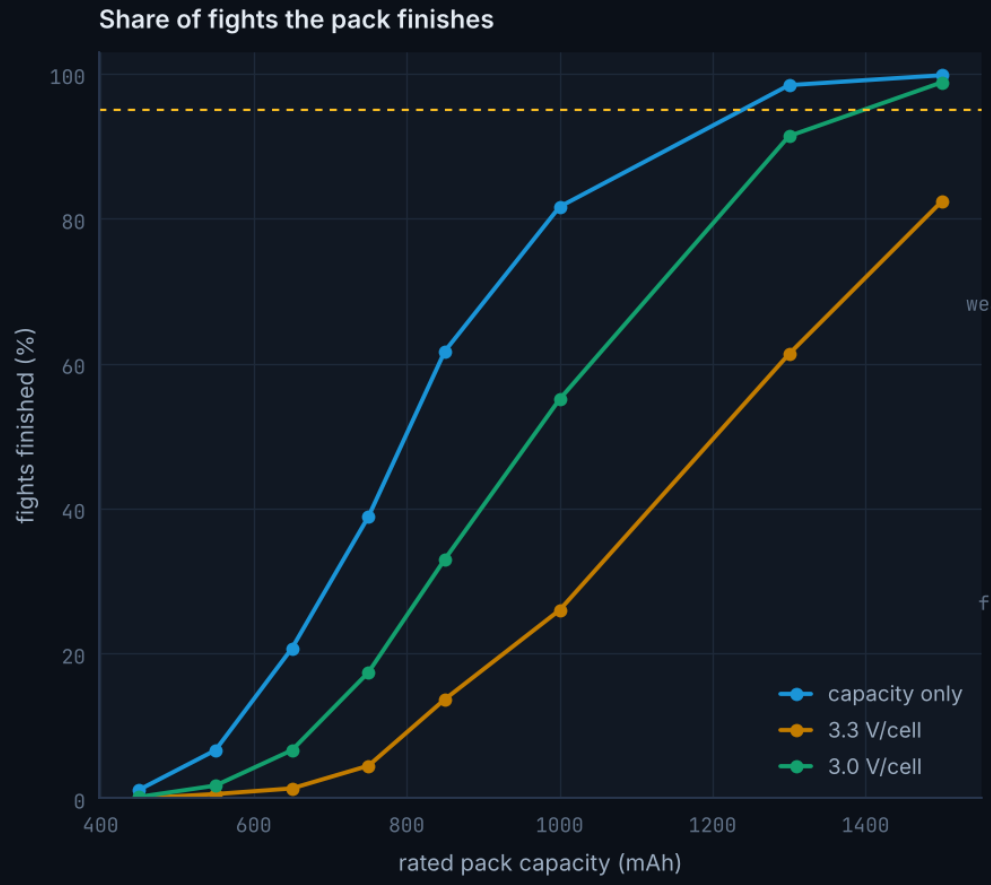


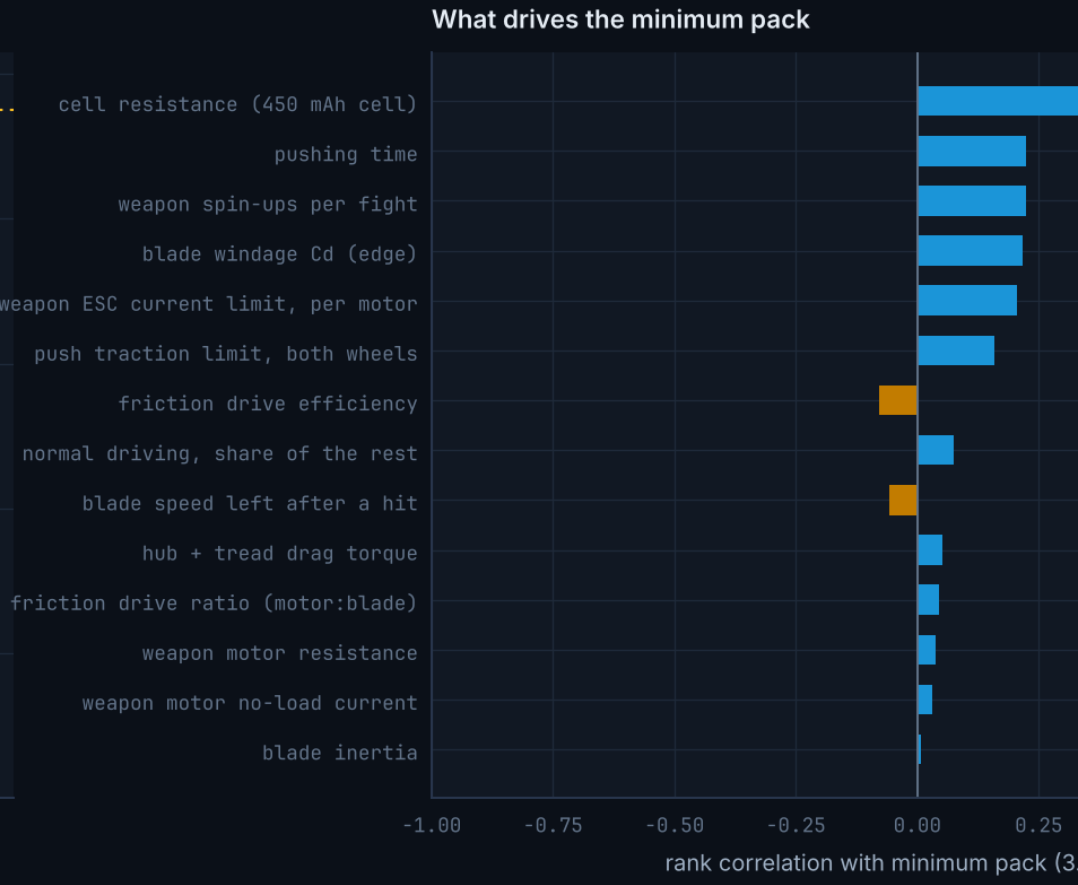
Battery over a 180 s fight: VERIFIED | PASS 14 WARN 2 FAIL 0 SKIP 0 | 51 runs, 7.0 s



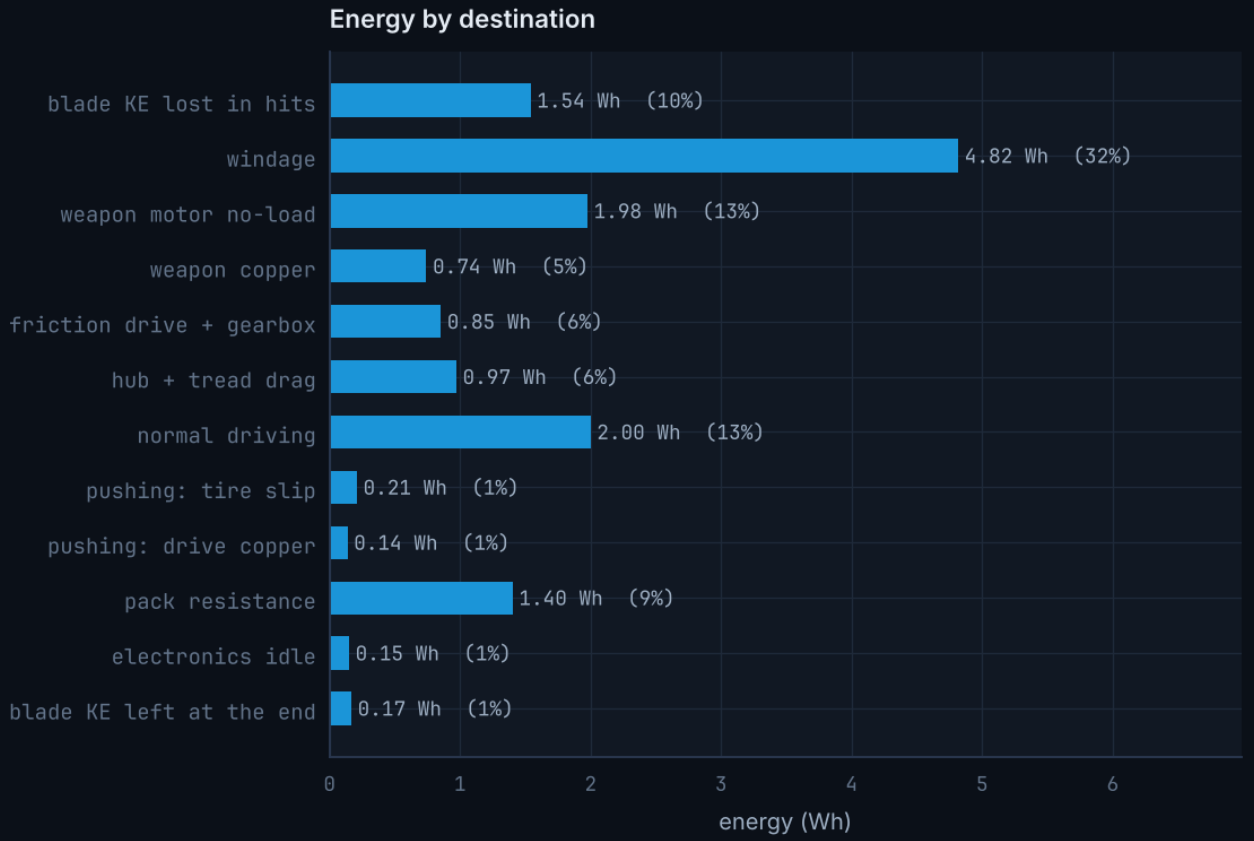


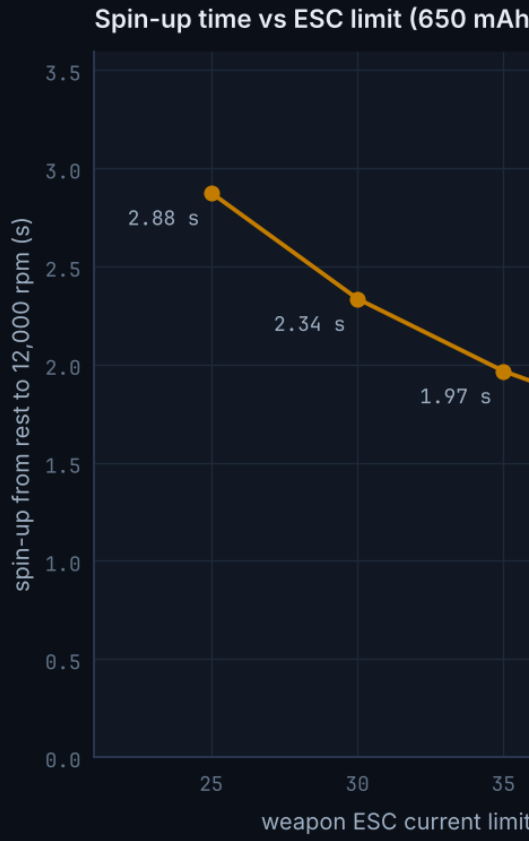
White bars: 95th percentile of the minimum pack for that spin-up count. Dashed lines: common 6S packs (mAh).



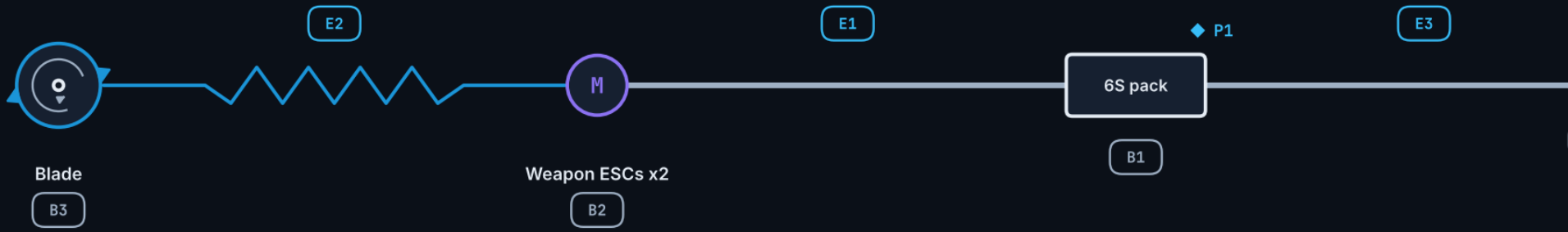


Blue: more of this needs a bigger pack. Orange: more of this needs a smaller pack. Dashed: 95% of fights.





Spin-up is current-limited the whole way: back-EMF at 12,000 rpm blade speed is about half the pack voltage.



KEY

spring · spring + slider · contact · damper · rigid link · plastic hinge · probe · E1 see table

PARAMETERS

B1 6S pack
OCV(SOC) + R0 + R1||C1
R0 scales with 1/capacity

B2 Weapon ESCs x2
2807 1350 KV
current limit 25-45 A

B3 Blade
J 7.6-8.8e-4 kg m^2
12,000 rpm target

B4 Drive ESCs x2
Compact V2 (1806 2300 KV)
20 A limit

E1 Bus
constant-power load
src: ecm_constant_power

E2 Friction drive
G 1.25-1.5, eta 0.75-0.95
src: esc_supply

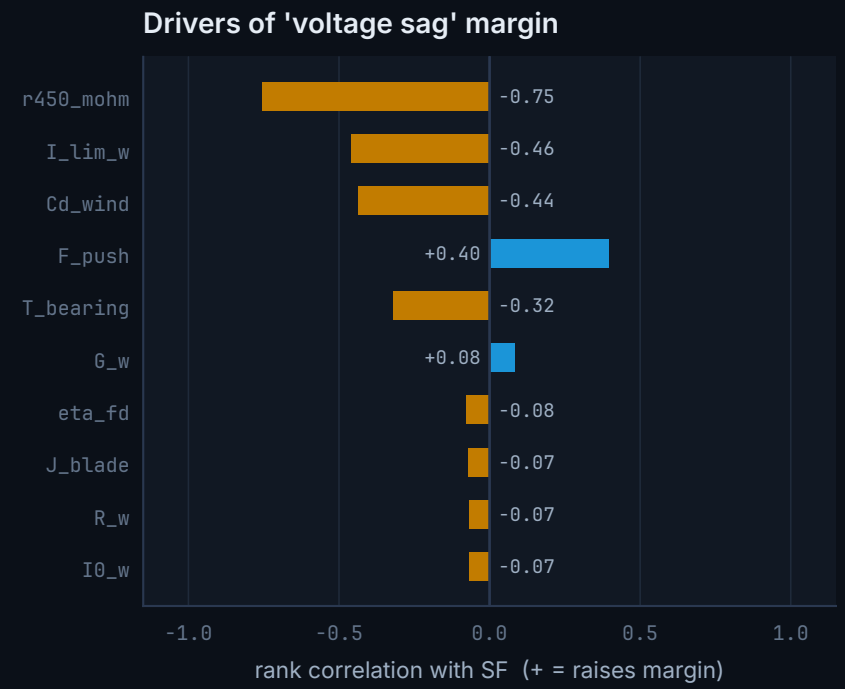
E3 Bus
drive power trace / push
src: motor_current

E4 Tires
push traction 8-150 N
src: coulomb_tanh

P1 charge used, loaded voltage
measured on 6S pack
src: failure check

BATTERY FOR A 180 S FIGHT

What to design around: drivers of the governing margin and the design-variable sweep



SIMBLOCKS // VERIFICATION

2026-10-16 08:03Z

BATTERY FOR A 180 S FIGHT

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 0.000867 s / dt 0.0001 s = 8.7x
dt_convergence	● WARN	Wh: err 9.1e-05% (order -0.53); mAh_used: err 9.8e-05% (order -0.55); v_min_cell: err 0.00091% (order 0.7); I_peak: err 0.0059% (order 0.96); t_fir...
energy_ledger	● PASS	worst closure residual 0.41% across 3 dt levels (tol 5%)
limit:idle only: 3 W x 180 s = 0.15 Wh	● PASS	matches expected limit
knockout:no blade windage	● PASS	Wh -26.8%, mAh_used -27.7%, v_min_cell +2.04%, I_peak -0.978%, t_first_full -3.48%
knockout:no weapon hold (governor off)	● PASS	Wh -41.5%, mAh_used -42.6%, v_min_cell +4.03%, I_peak -1.46%, t_first_full +0%
knockout:no pushing	● PASS	Wh -26.5%, mAh_used -27.4%, v_min_cell +3.41%, I_peak -4.02%, t_first_full +0%
knockout:no normal driving	● PASS	Wh -8.82%, mAh_used -9.17%, v_min_cell +3.8%, I_peak -14.3%, t_first_full +0%
knockout:no hits (one spin-up only)	● PASS	Wh -13.3%, mAh_used -13.8%, v_min_cell +4.69%, I_peak -27.6%, t_first_full +0%
knockout:no pack recovery circuit (R1 = 0)	● PASS	Wh -0.52%, mAh_used -0.542%, v_min_cell +1.25%, I_peak -1.24%, t_first_full +0%
sensitivity	● PASS	10 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	20 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	charge used: nominal SF 1.87, P(fail) 0%; voltage sag: nominal SF 1.04, P(fail) 20%